ARMSTRONG

DZ-104

Dual Column Touchscreen Tensile Testing Machine



Product illustration. Confirm the supplied configuration and accessories in the quotation.

MODEL	CAPACITY / CONFIGURATION
DZ-104	2,000 kgf / approximately 19.61 kN

Servo-driven dual-column test system with TS7003L touchscreen. The controller shows live force and displacement, captures peak readings and prints test data on the integrated micro-printer.

Supplied and supported by Hung Ta Instrument.

Applications and specifications

For mechanical testing of rubber, plastics, plastic film, wire, cable, textiles, waterproofing and nonwoven materials, and suitable metal wire, foil, sheet and bar specimens. Use the correct grips and fixtures for tensile, compression, shear, tear, adhesion and 90/180-degree peel testing.

Parameter	Specification
Rated test load	2,000 kgf (approximately 19.61 kN)
Frame and drive	Dual-column frame; servo motor
Control	TS7003L touchscreen measurement and control system
Load resolution	1/25,000
Load accuracy	+/-1% as stated in the source; verify the configured range
Maximum travel	800 mm excluding fixtures
Test speed	1 to 500 mm/min; touchscreen input
Data sampling	10 Hz
Display	Current test value and peak value; built-in micro-printer
Selectable force units	kgf, N and lbf; available units depend on controller configuration
Machine dimensions	1010 x 490 x 1730 mm
Machine weight	Approx. 150 kg
Supply	Single phase, 220 V; technical table lists 6 A or as specified. Confirm the nameplate.

For quotation, provide the specimen material and size, test method, required force range, fixtures and report requirements. Warranty and service are governed by the Hung Ta quotation.

Technical reference

Operating and software guidance. Illustrations may show example readings or a different software revision. Use the instructions for the delivered unit.

Safety and specimen preparation

Operate on a stable, ventilated surface. Follow the installed warning signs, use protective earthing and match the supply to the nameplate. High-voltage areas and internal wiring are for trained service personnel only.

Select specimens from the same material batch and cut them according to the test standard. Check both travel limits and fixture fasteners. Start with low jog speed, align the grips and secure the specimen before zeroing the controller.

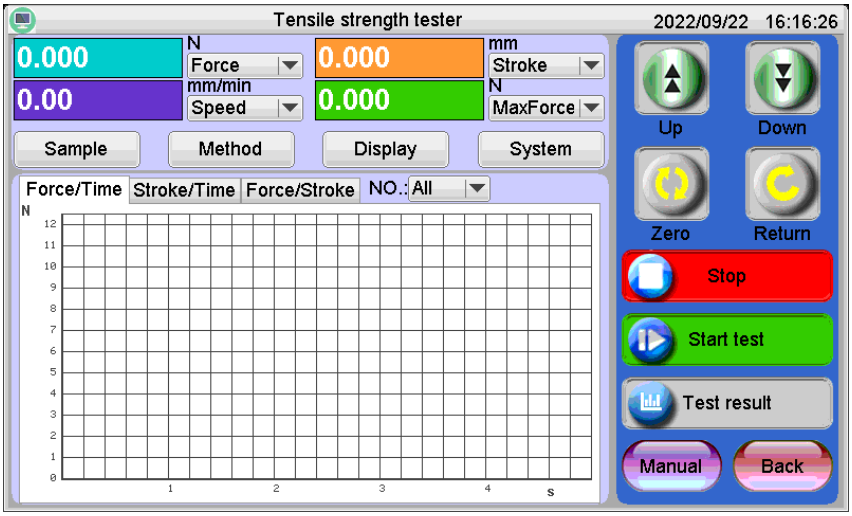
Setup and verification

The touchscreen uses force and displacement feedback and reads the upper and lower travel limits. Contact Hung Ta for commissioning, wiring, limit checks and calibration for the supplied configuration.

Test workflow and controls

Select the English interface, then set specimen data, test plan and display settings. Confirm the travel limits and specimen position, choose the force unit, zero readings and start the test. Watch the live and peak values and the force/time, travel/time or force/travel curve.

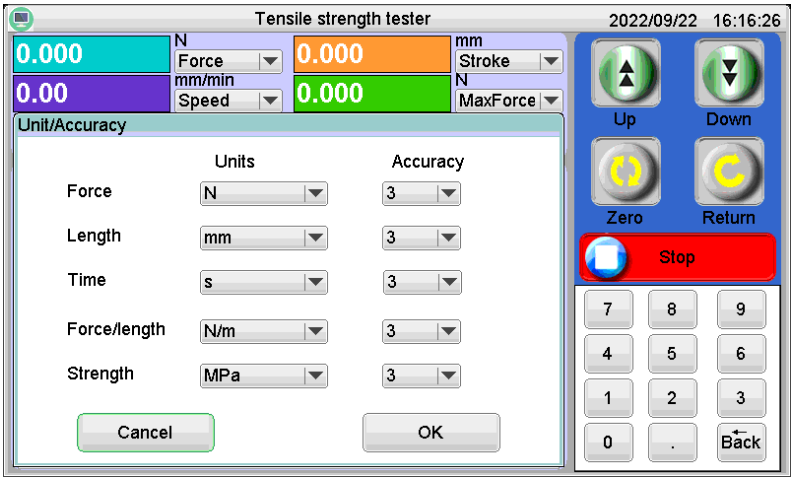
Start runs the configured test; Stop halts movement. Zero resets readings; Up and Down jog the crosshead; Return performs the configured return motion. Manual control allows a specified jog speed. After testing, remove the specimen safely and clean the machine.



Main test screen

Units and specimen data

Set force, length, time, force per unit length and stress units independently. Examples include N, kN, kgf, lbf, mm, in, seconds, minutes and MPa. Decimal precision can be configured from zero to three places. Enter specimen shape, gauge length, width, thickness or area as required for strength calculations.



Units and decimal places

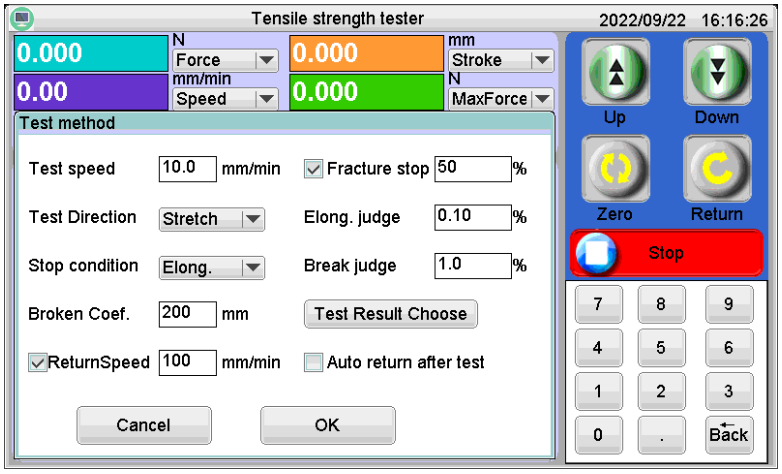


Specimen dimensions and area

Test plan and display settings

Set test and return speeds, tension or compression direction, break/deformation/force stop condition, break ratio and initial judgment thresholds. Enable return after testing when suitable. Compression holding settings include target force, hold time, deceleration and the start coefficient.

The source examples show an 85% break ratio and a 1% full-range break-detection threshold. These are examples, not prescribed settings for every test. Set sign conventions, zero-before-test options, curve printing, units and display precision as required.



Test plan and stopping conditions



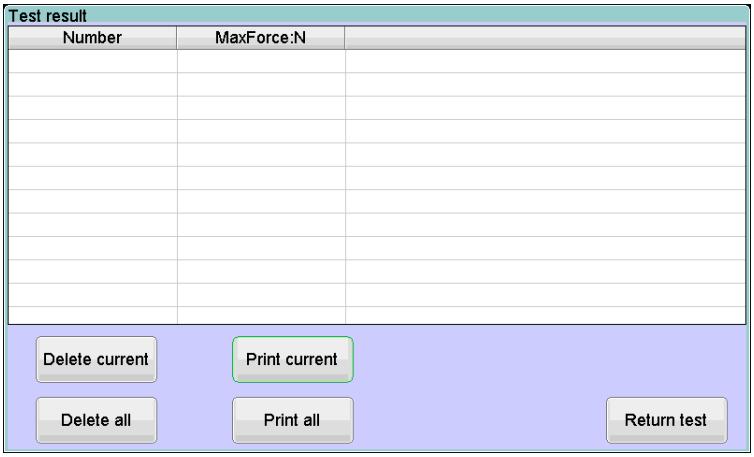
Force and holding settings



Display options

Test results and printout

Use Test Result to review individual or stored tests. Depending on the selected method, results may include peak force, deformation at peak, tensile strength, tear strength and peel strength. Print the current result or the stored results.



Stored results and print controls

Care and troubleshooting

Keep the frame stable and ventilated, keep the instrument grounded and do not move it during operation. Keep the machine clean; wipe the controller with a dry cloth and maintain mechanical lubrication as specified.

If the display is blank, check the external supply and cable. If motion fails, or motion continues beyond a limit, stop using the machine and contact Hung Ta for controller or limit-switch diagnosis. Do not bypass the limits or open live panels.

Service and support

For technical support, provide the ARMSTRONG model, serial number and a description of the issue to Hung Ta Instrument.